Surgical punctal occlusion with a high heat-energy releasing cautery device for severe dry eye with recurrent punctal plug extrusion.

PURPOSE: To report the rate of recanalization and the efficacy of punctal occlusion surgery with a high heat-energy-releasing cautery device in patients with severe dry eye disease and recurrent punctal plug extrusion. DESIGN: Prospective, interventional case series. METHODS: Seventy puncta from 44 eyes of 28 dry eye patients underwent punctal occlusion with thermal cautery. All patients had a history of recurrent punctal plug extrusion. A high heat-energy-releasing thermal cautery device (Optemp II V; Alcon Japan) was used for punctal occlusion surgery. Symptom scores, best-corrected visual acuity, fluorescein staining score, rose bengal staining score, tear film break-up time, and Schirmer test values were compared before and 3 months after the surgery. Rate of punctal recanalization also was examined. RESULTS: Three months after surgical cauterization, symptom score decreased from 3.9 ± 0.23 to 0.56 ± 0.84 ($P < .0001$). Logarithm of the minimal angle of resolution best-corrected visual acuity improved from 0.11 ± 0.30 to 0.013 ± 0.22 ($P = .003$). Fluorescein staining score, rose bengal staining score, tear film break-up time, and the Schirmer test value also improved significantly after the surgery. Only 1 of 70 puncta recanalized after thermal cauterization (1.4%). CONCLUSIONS: Punctal occlusion with the high heat-energy-releasing cautery device not only was associated with a low recanalization rate, but also with improvements in ocular surface wetness and better visual acuity.